GMC LANKA CREATIONS

Performance Test Report

SENG 34213 – System Development Project

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Version history

Version No.	Date	Author	Summary of Changes
01	2026/10/24	Team	Load test of release 0.1.1

CHAPTER 1 OBJECTIVES

The performance tests verify the measurable non-functional requirements of the SRS that relate to response time and load:

Table 1.1: Requirements under test

Requirement	Acceptance criterion
NFR001	95% of search queries return results within 2 seconds under normal load
NFR002	The catalogue page loads within 3 seconds for up to 50 concurrent users
NFR012	The system supports at least 100 concurrent users without performance degradation

CHAPTER 2 TEST ENVIRONMENT AND METHOD

Table 2.1: Test environment

Item	Value
Application version	Backend 0.1.2-SNAPSHOT (release 0.1.1 code), Spring Boot 4.1.1, Java 21
Machine	Intel Core i5-1135G7 @ 2.40 GHz, 8 logical cores, 16 GB RAM, Windows 11
Database	Local MySQL server on the same machine
Data set	1 active product, 5 categories
Tool	Python 3.13 load script (thread pool, standard library), read-only GET requests
Date	2026-10-24 18:33:39

The script first sends 20 warm-up requests so that just-in-time compilation and the connection pool are not measured. Each scenario then runs a fixed number of concurrent virtual users, each sending requests back to back. Response time is measured from sending the request to receiving the full body. Percentiles are calculated over all requests in the scenario. Only public, read-only endpoints were used, so the test does not change any data.

Table 2.2: Scenarios

Scenario	Endpoints	Users	Requests
NFR001 search (GET /api/products?search=)	GET /api/products?search={wood, teak, cnc, panel, table, a}	20	500
NFR002 catalogue page (products + categories)	GET /api/products, /api/products/categories, /api/products?categoryId={id}	50	1000
NFR012 mixed browsing, 100 users	Product list, categories, product details and product reviews	100	1000

CHAPTER 3 RESULTS

Table 3.1: Response times in milliseconds

Scenario	Throughput (req/s)	Mean	p50	p95	p99	Max	Errors
NFR001 search	157.9	122.3	111.9	212.4	267.0	341.8	0 (0.0%)
NFR002 catalogue page	155.2	310.8	308.0	500.9	637.1	800.5	0 (0.0%)
NFR012 mixed browsing, 100 users	148.0	615.3	619.1	912.9	1491.3	2217.1	0 (0.0%)

Table 3.2: Verdict against requirements

Requirement	Target	Measured	Verdict
NFR001	p95 ≤ 2,000 ms	p95 = 212 ms	Pass
NFR002	≤ 3,000 ms at 50 users	p95 = 501 ms, max = 800 ms	Pass
NFR012	100 users, no errors	0 errors, p95 = 913 ms	Pass

CHAPTER 4 ANALYSIS AND RECOMMENDATIONS

- Throughput stayed between 148 and 158 requests per second while the number of users grew from 20 to 100, so additional users mainly increased waiting time: the median rose from 112 ms to 619 ms. The server was saturated rather than failing, and no request returned an error.
- The 100-user scenario had a maximum of 2217 ms; this is still within the 3-second target but shows that the 'no degradation' wording of NFR012 is only met relative to the targets of NFR001 and NFR002.
- The catalogue endpoint returns every product without server-side pagination. With a realistic catalogue of hundreds of products, response size and time would grow; server-side pagination and HTTP caching of the category list are recommended before production use.
- The production host (a small EC2 instance with 1 GB of memory and MySQL on the same host) has fewer resources than the test machine. The test should be repeated against the staging environment with a full product catalogue before go-live.